# Navigating AI in evidence synthesis

## AI tools across the review workflow

Luke Baxter

---

Map and presentation

lukebax.github.io/ai-review-tools/

# Stages of a systematic review

1 Protocol

2 Search

3 Screening for relevance

4 Quality assessment

5 Data extraction

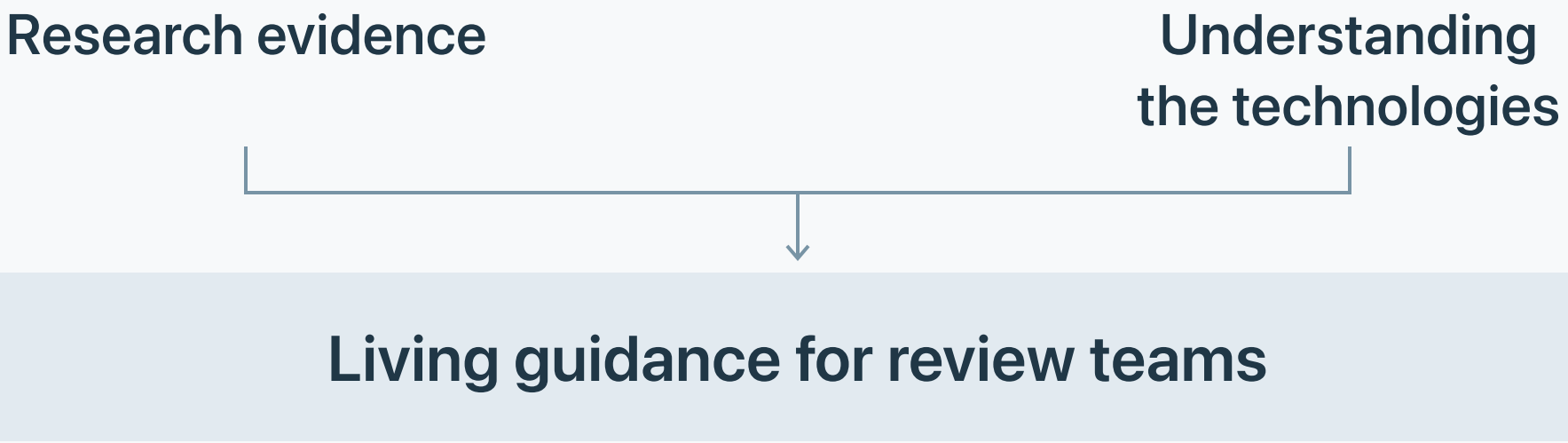
6 Data synthesis

7 Certainty of evidence

8 Manuscript

# RAISE provides guidance for using AI in reviews

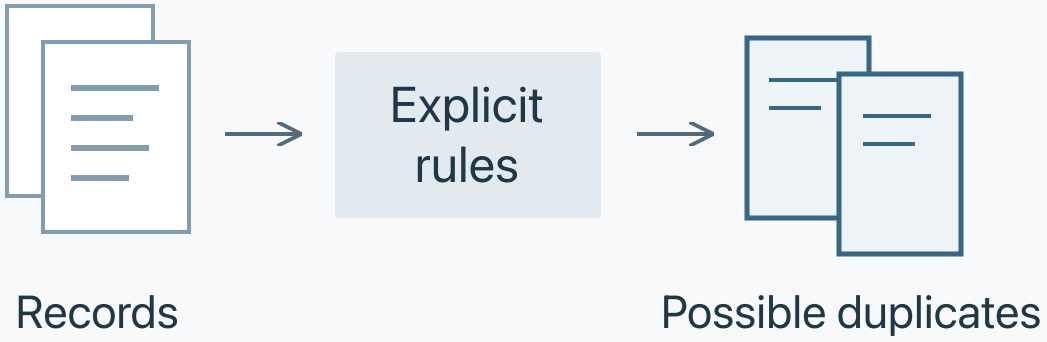
Responsible use of AI in evidence synthesis




Developed by researchers working in evidence synthesis

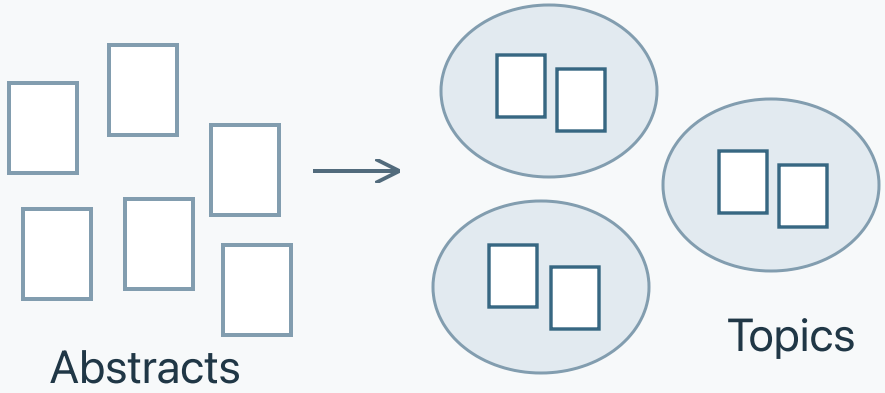
# Explicit rules and patterns in unlabelled data

## Rule-based algorithms




People define the rules used to compare records.

## Unsupervised machine learning

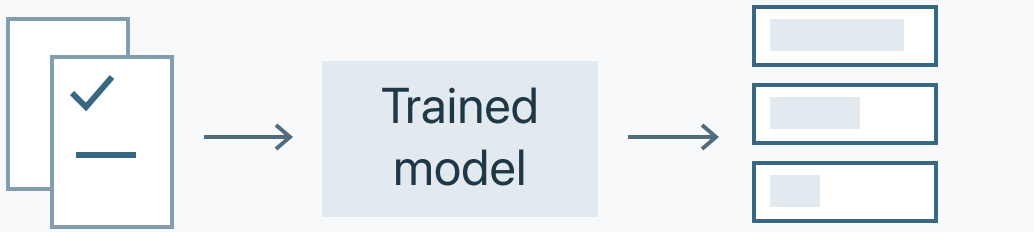



Patterns emerge without supplied labels.

Results depend on the records and the method.

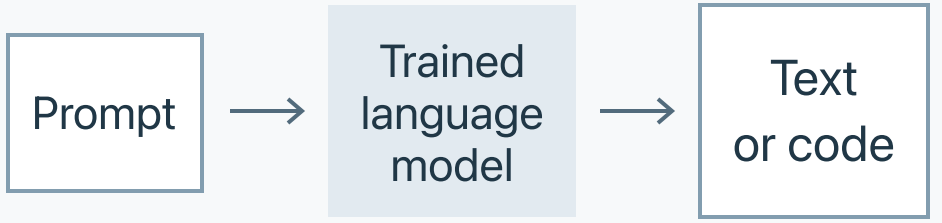
# Learning from labelled examples and generating language

## Supervised machine learning




Already trained or trained during the review

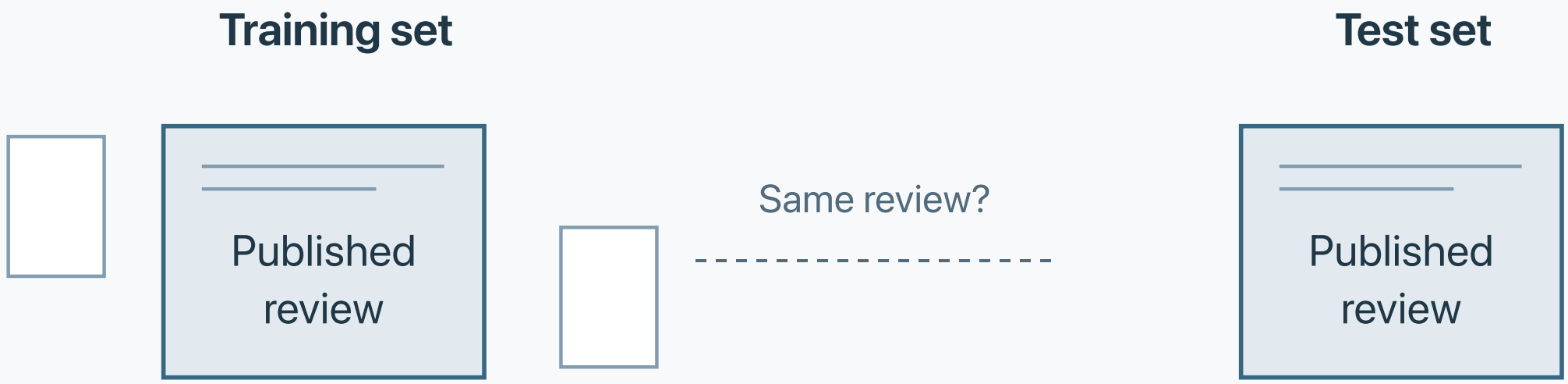
## Generative LLMs




Outputs can vary.
Fluent output can be wrong.

# Evaluation needs independence from development

Possible training–test overlap: a hypothetical example




Apparent performance may overstate performance on new material.

Incomplete disclosure makes overlap harder to establish.

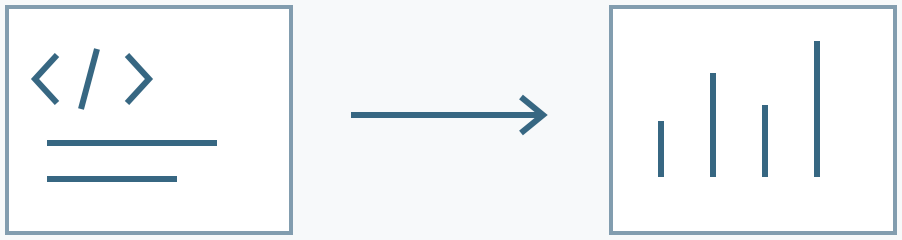
# Recommendations apply to a particular use

| | |
|---|---|
| **Acceptable for use** | Use outputs, accounting for limitations and bias |
| **Human verification required** | Check outputs and correct errors |
| **Requires validation within the review** | Test performance against a prespecified threshold and suitable benchmark |
| **Exploratory and supplementary use** | Explore or supplement established methods |
| **Not acceptable for use** | Do not rely on outputs for this task |

# Verification and validation ask different questions

## Verification

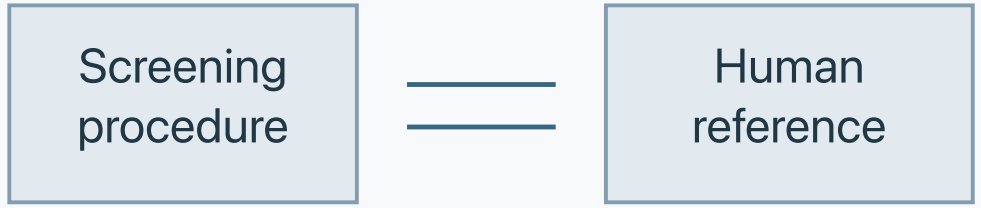
Check the output




Does it carry out the intended analysis?

## Validation

Evaluate performance



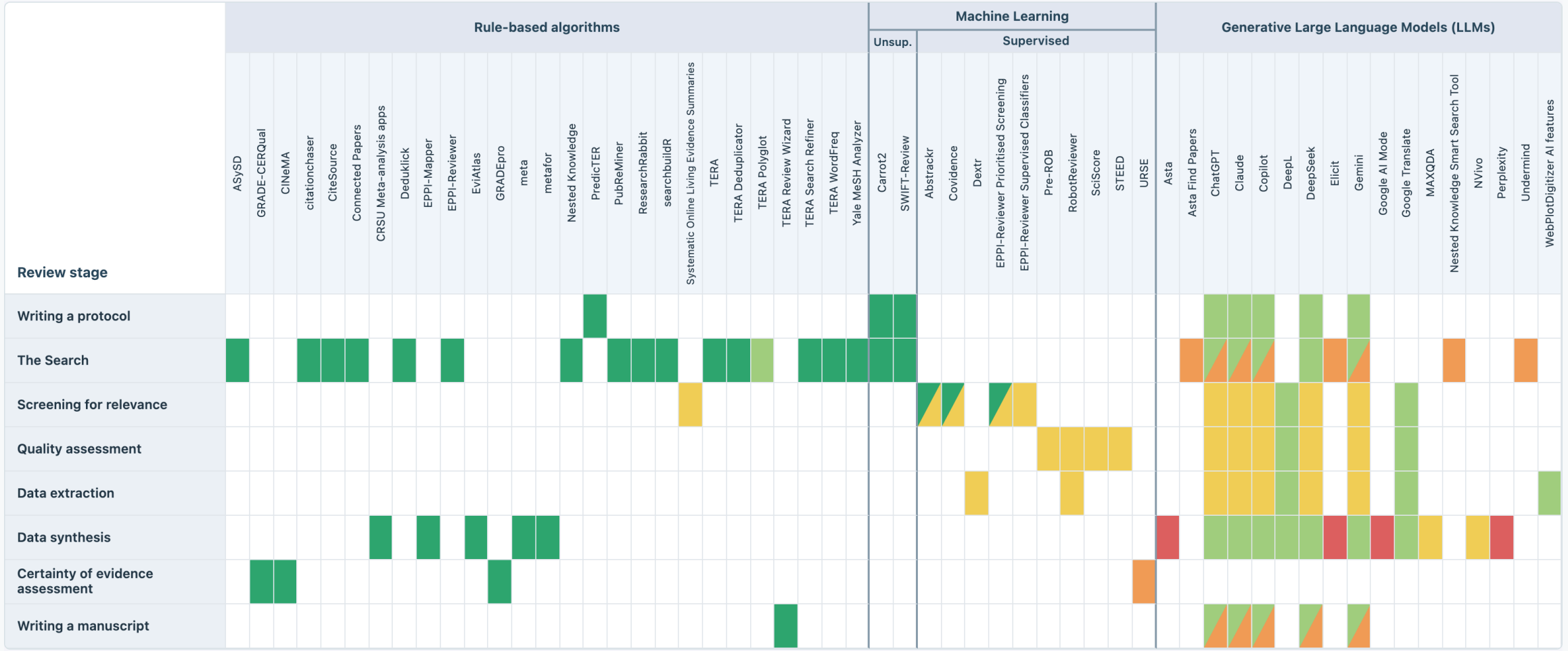

Recall: proportion of eligible records found

Applicable prior evidence may suffice. Follow any within-review requirement.

Suitable sampling or stopping methods can help, but assessment has a workload cost.

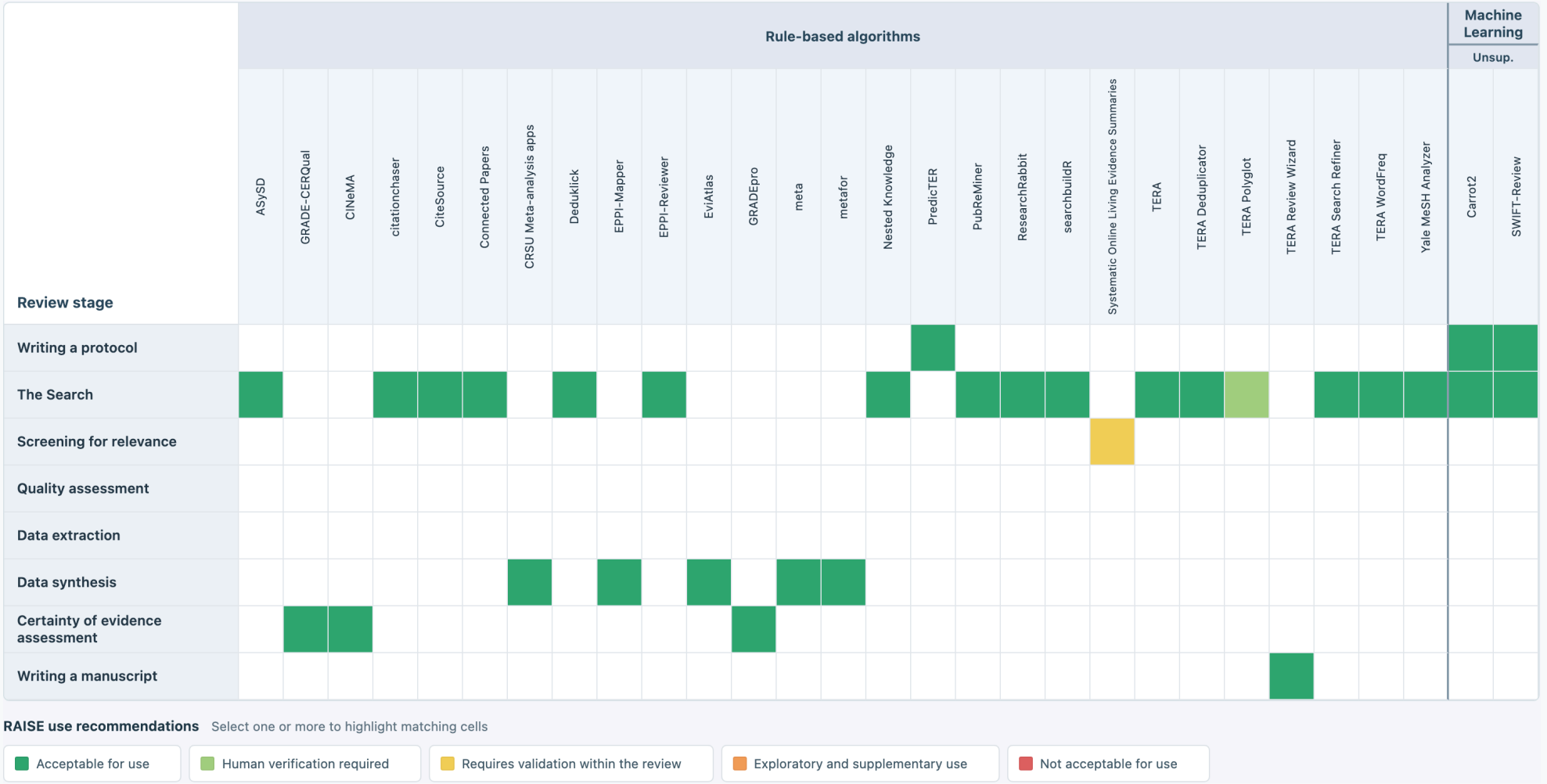
A workflow may need both.

# AI tools across the review stages



# Rule-based algorithms and unsupervised machine learning

## Most of these mapped uses are acceptable



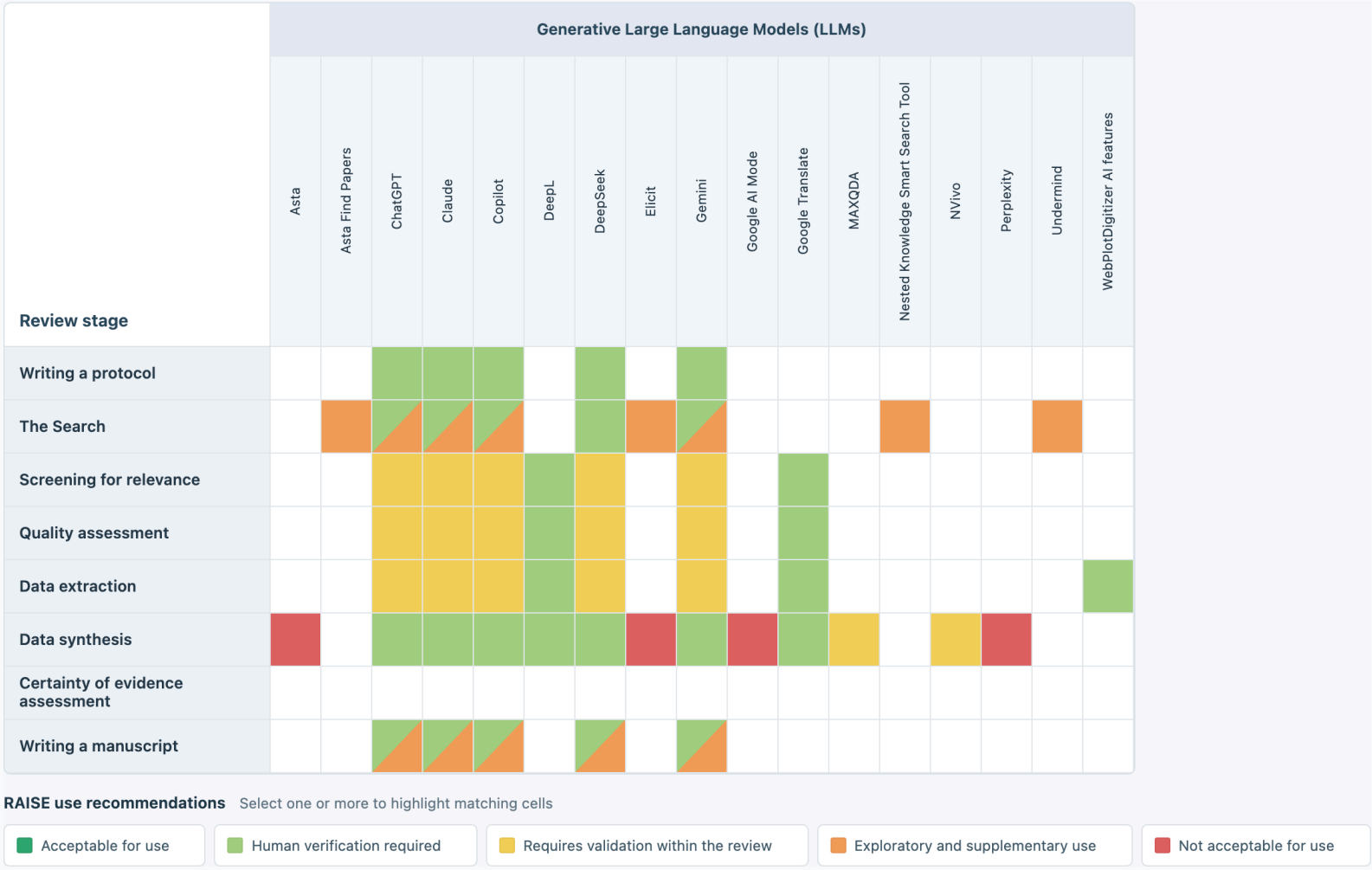
# Supervised machine learning

## Most mapped uses require validation within the review



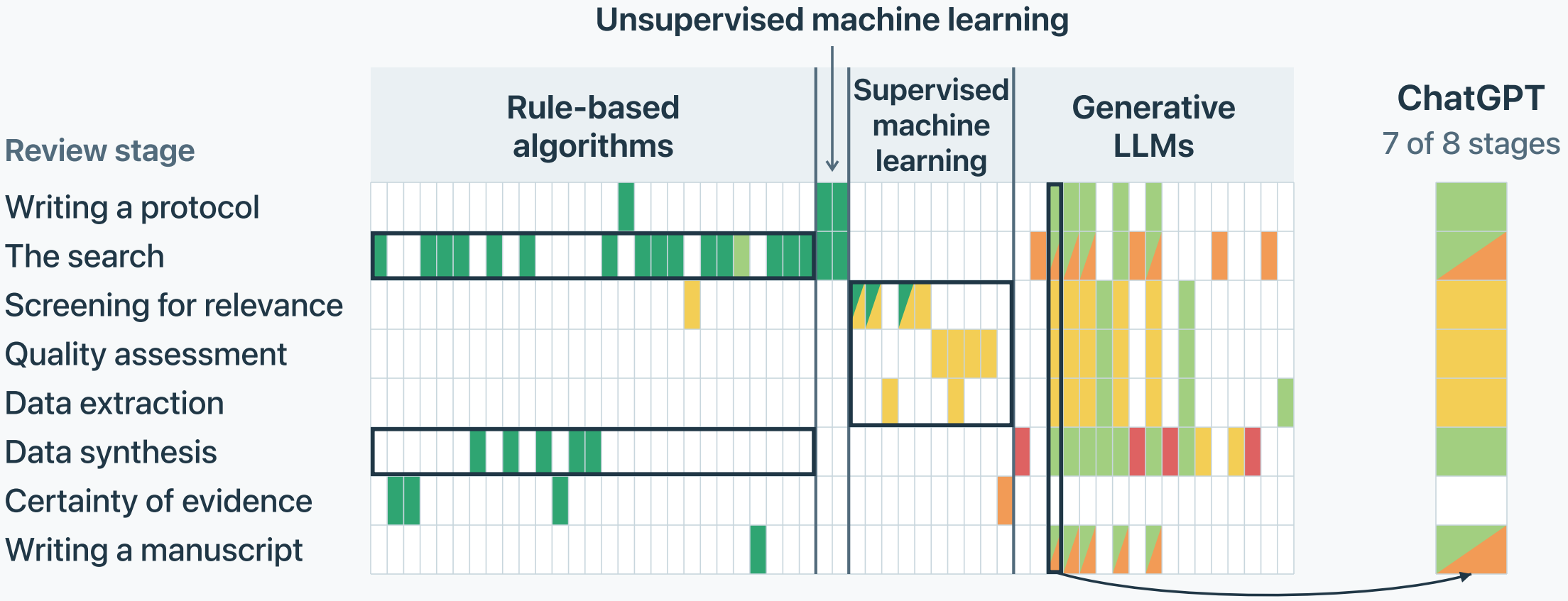
# Generative large language models (LLMs)

## The mapped uses carry conditions or restrictions



# Tools differ in where and how broadly they are used

Stage clusters in the overview, with ChatGPT enlarged alongside




Rule-based
Search and synthesis

Supervised
Screening, quality assessment, extraction

ChatGPT
Mapped in 7 of 8 stages

# Prioritised screening with and without early stopping

Stopping early changes reliance on the ranking

**Abstrackr**



**Covidence**



**EPPI-Reviewer Prioritised Screening**



## Screen every record

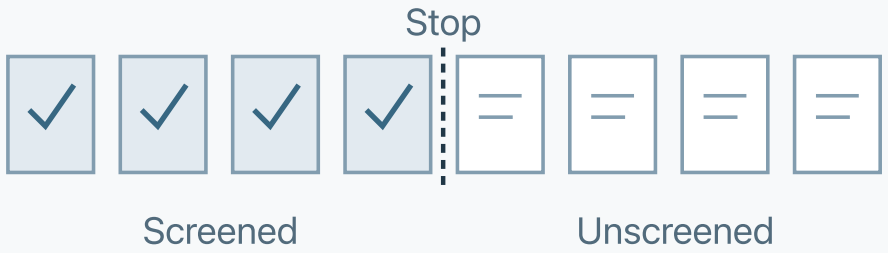


Every record is screened

**Acceptable for use**

A poor ranking does not leave records unseen.

## Stop early




**Requires validation within the review**

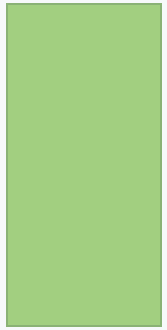
Assess the screening and stopping procedure.

The ranking can be identical. What changes is how much we rely on it.

# Two different uses during screening

Screening for relevance

## Google Translate

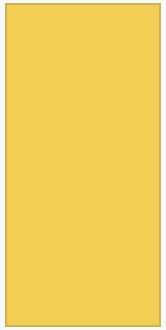


Translation support

**Human verification required**

Check the translation against the original.

## Google’s Gemini



Relevance judgements

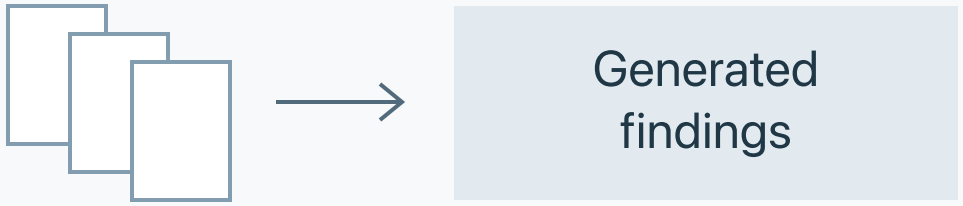
**Requires validation within the review**

Evaluate the screening procedure’s performance.

Selected screening cells from the map.

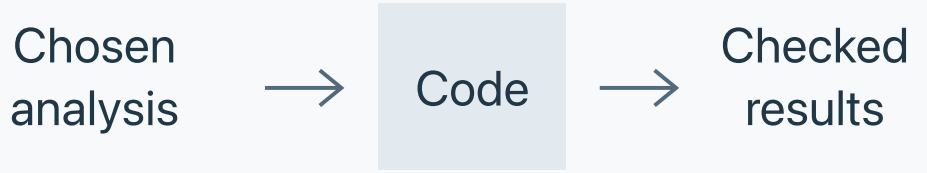
# Synthesising findings differs from drafting analysis code

## LLM-derived synthesis across studies




**Not acceptable for use in a formal review**

## LLM-drafted code for a chosen analysis

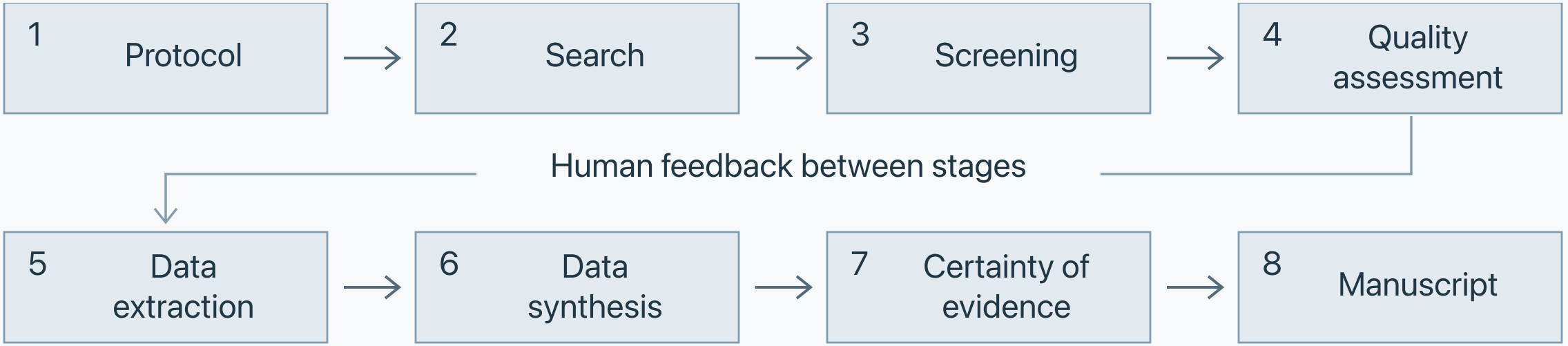



**Human verification required**

Qualitative coding or themes: validation and verification.
The boundary with delegated synthesis is not fully defined.

# RAISE also restricts complete agentic reviews

**Complete agentic review: not acceptable for use**




Human feedback does not remove this restriction.

Individual tasks need separate assessment.

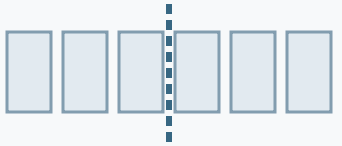
# Is this tool suitable for my review?

Prioritised screening with early stopping

1

## What is the purpose of the tool?

Assess ranking plus the stopping procedure.



2

## Where have the training, testing and validation data come from?

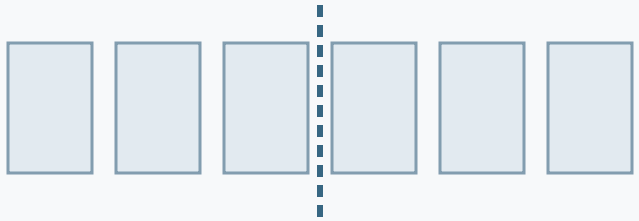
Review-team labels: quality and selection.
Method evaluations: relevance to this review.



# Validation and performance

Prioritised screening with early stopping

## 3 Is the tool validated and performing sufficiently well for use?



Stopping procedure

Compare performance

Human-screened reference

Evidence for the stopping procedure

Set required performance in advance.
Establish whether it is met within this review.

An appropriate human-screened reference

# Team, resources and documentation

Prioritised screening with early stopping

4 **Does the team have the expertise and resource to use the AI tool effectively, and is the usability acceptable?**

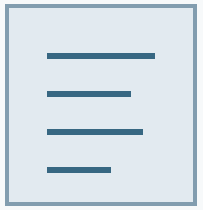
Skills, assessment time and a workable process

5 **Is there sufficient transparency around licenses and other available documentation?**

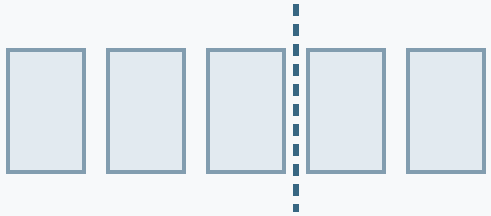
Instructions, access, data handling, version and settings

Missing stopping evidence must be resolved before reliance.

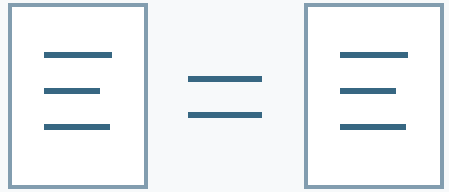
# Choosing a suitable use of AI



Tool and task



Reliance on outputs



Evidence and checks

Map and presentation

**lukebax.github.io/ai-review-tools/**